DISPLAYS AND ALERTS

ENP-0112

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Dr. Divya C. Chandra

TOPICS

- Visual displays
- Auditory displays
- Alerts in aviation
- Collision avoidance
- Implementation and testing

VISUAL DISPLAYS

VISUAL DISPLAY DESIGN CONSIDERATIONS

- Display size
- Screen resolution
- Off-axis viewing
- Luminance
- Brightness
- Font
  - Size (in visual angle)
  - Readability (e.g., serifs and spacing)
- Field of view
- Vibrations
- Dark adaptation
- Density
- Clutter
- Color and contrast
- Highlighting
- Grouping
- Graphical vs. text representation of data
- Analog vs. Digital
- Redundant coding

See also, Proctor & Van Zandt (Reading List)

VISUAL DISPLAY DESIGN PRINCIPLES

- Principle of Information Need
- Proximity Compatibility
- Redundancy
- Principle of Pictorial Realism
- Principle of the Moving Part
- Predictive Aiding

Wickens et al., 2004

AUDITORY DISPLAYS

TYPES OF AUDITORY DISPLAYS

- Symbolic
  - Map sounds to meanings
    - e.g., emergency preparedness signal, mail
- Non-speech
  - Tones, combinations of frequencies, patterns
  - Can vary in time
- Speech
  - Can be used in combination with non-speech
  - Can convey more complex information, if the person is alert

Aural warnings test

AUDITORY ALARMS

- Primarily used for emergency alarms and warning signals

  - ***True alarms***: alarms sounded when intended
  - ***False alarm***: sounds, but does not need to
  - ***Missed alarm***: should have sounded, but did not
  - ***Nuisance alarm***: sounds as intended, but when the alert is not really needed for operations

ALARM INCIDENTS IN CLINICAL CARE

- False alarms problematic (81% respondents)
- Nuisance alarms disruptive to patient care (77% respondents)
- Distrust alarms and disable devices (78% respondents)

nursing.advanceweb.com

Korniewicz D, Clark T, David Y. A National Online Survey on the Effectiveness of Clinical Alarms. American Journal of Critical Care Management 17(1): 36-41, 2008.

AUDITORY ALARM ISSUES IN AIR TRAFFIC CONTROL

0 (not a problem) to
10 (severe problem)

Table 2. Mean Ratings of Items by Terminal AT Participants

Problem	Mean	St Dev
Alarms are easily confused (because they sound alike)	6.7	3.18
Alarms go off too frequently, especially false alarms	6.4	3.57
Alarms are annoying	6.3	2.53
It is difficult to locate the source of the alarms	5.3	3.26
Too many alarms go off at the same time	5.2	3.33
There are too many alarms for a person to learn the meaning of each alarm	4.9	3.41
Alarms sound more urgent than they should or sound less urgent than they should	4.8	3.16
Alarms are too loud	4.5	2.91
Alarms disrupt thought	4.4	3.66
Alarms can be masked (difficult to hear over background noise)	4.1	3.43
Alarms interfere with voice communications	4.0	2.87
Alarms startle the user	3.8	2.77
Some alarms which are visual would be better auditory and vice versa	3.1	2.84
There are not audio alarms in some situations where there should be audio alarms	2.1	2.29
Alarms go off so infrequently, that when they do go off, those hearing the alarms don't know their meaning	1.9	2.39

From Ahlstrom, 2003 DOT/FAA/CT-TN03/10

URGENCY

- Too loud is ineffective; most effective way to increase urgency is to increase speed of warning
- Urgency categories and central warning design
  - See Title 14 Code of Federal Regulations Section 25.1322 Flightcrew alerting

Level of Urgency	Meaning	Visual Cue	Auditory Cue
Warning	Immediate action required	Red	Nonverbal to get attention, backup verbal message for more information
Caution	Immediate awareness and subsequent response	Amber or Yellow	Non verbal (optional verbal)
Advisory	Awareness and may require subsequent response	Any color other than red or green	n/a
Non-alert	Information only	Green, white, cyan, etc.	None (Quiet/Dark)

Edworthy, J. Loxley, S., and Dennis, I. (1991). Improving Auditory Warning Design: Relationship between Warning Sound Parameters and Perceived Urgency. *Human Factors* 33(2) pp. 205-231. https://doi.org/10.1177/001872089103300206

AUDITORY VS. VISUAL DISPLAYS

	Auditory	Visual
Reception	Omnidirectional	Must be in field of view (Attention & selection)
Speed	Fastest	Slowest
Order	Difficult	Easy
Urgency	Easy	Difficult
Noise	Not affected by visual	Not affected by audio
Symbolism	Melodious, linguistic	Pictorial, linguistic
Mobility	Most flexible	Some flexibility
Suitability	Time-dependent info	Space-dependent info

WHEN TO USE AUDIO OR VISUAL DISPLAYS

Auditory	Visual
Message is simple/short	Message is complex/long
Message won't be referred to later	Message will be referred to later
Message deals with time	Message deals with space
Immediate action required	Immediate action not required
Visual channel overburdened	Auditory systems overburdened
Brightness/Darkness problems	Noisy environment
Person is moving	Person is static
Communications with many people at the same time	*Communication with one (or few) individuals, or when info has to be distributed at different times*
When multiple complex data sources are monitored compared	*When high resolution of variables is required or when absolute (instead of relative) values are required*
When 3D aspect can be represented (spatialized sound)	*When sound would interfere with the task (e.g., speech) or be masked by the environment*

ALERTS IN AVIATION

CONSIDERATIONS FOR DESIGN OF AURAL WARNINGS

- Goal: *Accomplish intended function without disrupting user's ongoing tasks and information processing*
- What's the intended function?
  - Attract attention (alert), convey information, or both?
    - Cue to perform an action or indication that a defined threshold has been reached
    - Additional context/explanation (urgency, nature of the problem)
    - Voice messages both alert and inform
    - Avoid startle response (onset rate)
  - Convey urgency
    - Warning, caution, advisory

MORE DESIGN CONSIDERATIONS FOR AURAL WARNINGS

- Compliance
  - Intervention immediacy vs. intervention importance
- Effectiveness
  - Understandability +
- Learning time and retention
- Number of different warnings
  - More is less (potential for confusion)
  - Similar temporal patterns are confusable, as are single tones
  - How many is too many?

STILL MORE DESIGN CONSIDERATIONS FOR AURAL WARNINGS

- Background noise & prominence
  - Match warning frequencies and amplitudes to environment
  - Auditory masking
  - Ensure audibility without disrupting subsequent speech
  - Cascading alarms & cacophony
- Standardization
  - User expectations and prior learning
- Testing for comprehension and usability

COLLISION AVOIDANCE

PILOT TRAINING — “SEE AND AVOID”

- “Rules of the air” (right of way) in uncontrolled airspace (without ATC)
  - **Title 14 Code of Federal Regulations (CFR)** § 91.113 Right-of-way rules: Except water operations**.**
  - **Title 14 CFR** § 107.37 Operation near aircraft; right-of-way rules [For Unmanned aircraft]
- For example, from **§ 91.113**

(d) **Converging**. When aircraft of the same category are converging at approximately the same altitude (except head-on, or nearly so), the aircraft to the other's right has the right-of-way. If the aircraft are of different categories—

(1) A balloon has the right-of-way over any other category of aircraft;

(2) A glider has the right-of-way over powered aircraft.

(3) An airship has the right-of-way over all other powered aircraft, except for an aircraft towing or refueling other aircraft.

(4) An aircraft towing or refueling other aircraft has the right-of-way over all other powered aircraft.

(e) ***Approaching head-on.*** When aircraft are approaching each other head-on, or nearly so, each pilot of each aircraft shall alter course to the right.

- Self-announce position when landing and departing uncontrolled airports (without ATC)

SURVEILLANCE TECHNOLOGIES

- Primary and Secondary Surveillance Radar
  - Center (12 sec) vs. Terminal (5-sec) rotation rates
  - Position uncertainty if aircraft is turning or climbing/descending
- Automatic-Dependent Surveillance – Broadcast (ADS-B)
  - Out: own aircraft transmits position
  - In: own aircraft receives positions from other aircraft
  - 1-second refresh rate, position plus other data
- Airborne surveillance (TCAS)
  - Some version of TCAS is mandatory on most commercial aircraft
    - See for example, 14 CFR § 121.356 Collision avoidance system
  - Traffic display and vertical resolution advisories
  - Assumes 5-sec response time for pilot
- Airborne Collision Avoidance System – X (ACAS X family)
  - Incorporates neural networks
  - Xu version is for unmanned aerial vehicles, sXu for small unmanned aerial vehicles

Limitations of the technology affect alert design and logic.

INTEGRATED TRAFFIC DISPLAYS

Traffic alert and collision avoidance system (TCAS) displays may be integrated with a vertical speed indicator or primary flight display

TCAS display, photo by Paul Nelson

Figure 2-1. Resolution Advisory/Vertical Speed Indicator (round dial vertical speed indicator)

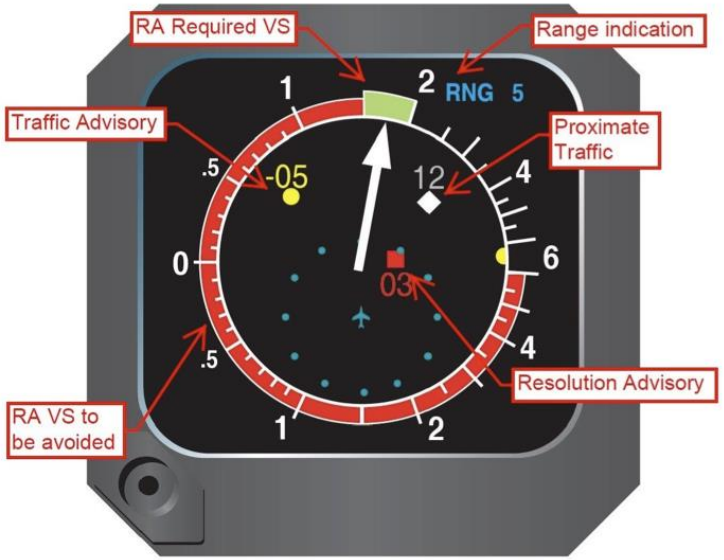
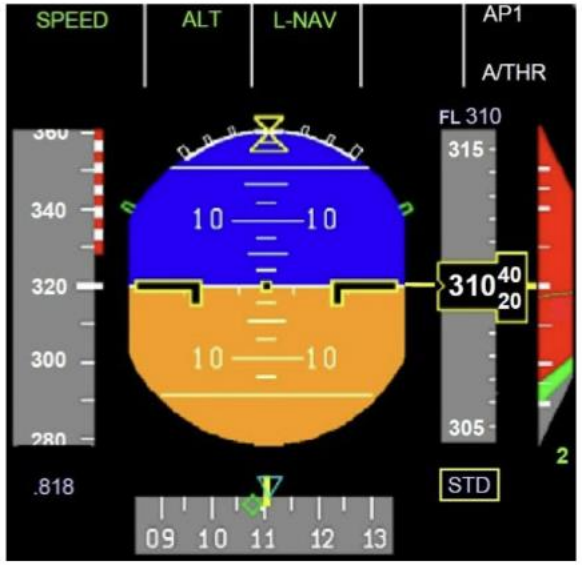


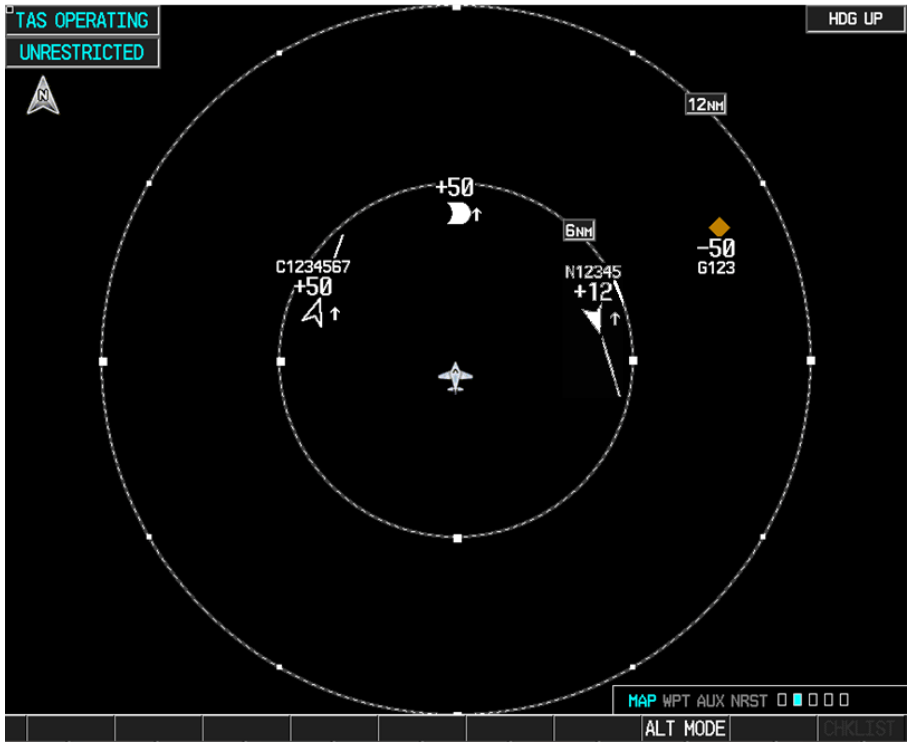
Figure 2-2. Integrated Vertical Speed Indicator Tape on a Primary Flight Display



Figures 2-1 and 2.2 from AC 90-120 (November 2024)

ADS-B TRAFFIC DISPLAY

Automatic-Dependent Surveillance-Broadcast (ADS-B) technology

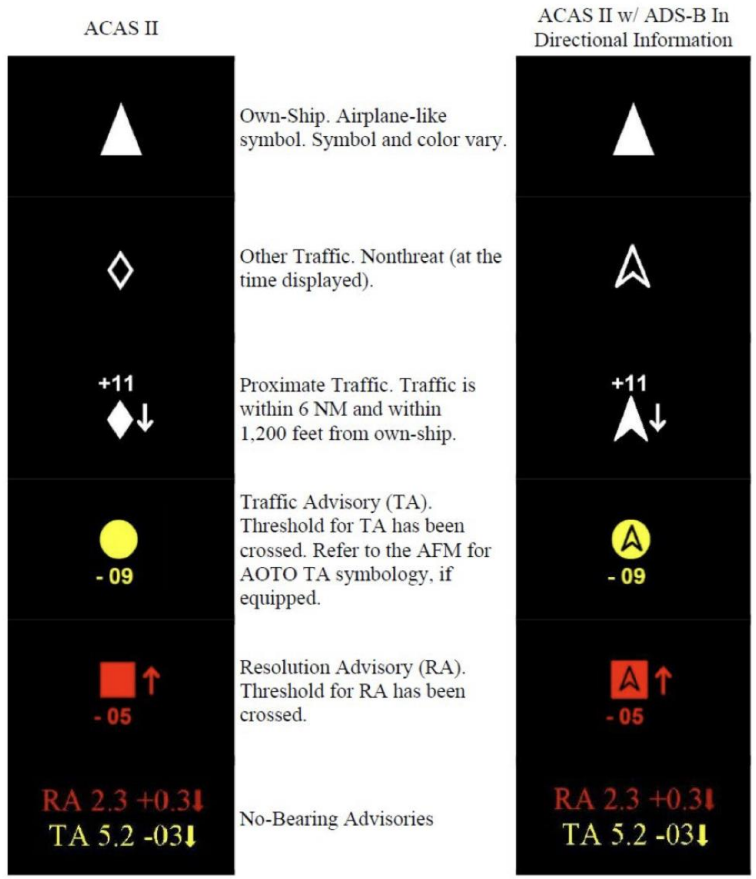


- New symbols can present more information
  - e.g., data quality, directionality
- How much information can be encoded visually in a traffic symbol without confusing the pilot?

Garmin G-1000 Cockpit Display of Traffic Information (CDTI) (RTCA SC-186, 2008)

TRAFFIC DISPLAY SYMBOLOGY EXAMPLES (AC 90-120)

Table 2-5. Traffic Display Symbology Examples



TCAS traffic display (left column) shows symbols:

- Traffic Alerts (TAs) [yellow circle]
- Resolution Advisories (RAs) [red square]
- Proximate traffic as either hollow or filled diamond
- Relative altitude and climb/descent information

And generates aural messages, such as:

- “Traffic, Traffic” (TA)
- “Climb, Climb” (RA)

ACAS II or ADS-B symbols in right column

TCAS AURAL WARNINGS

- TCAS test
  - Without and with audio
- Altitude tape version of TCAS command

What would you do if you only saw this display of vertical speed?

TCAS
TVSI/RVSI

IMPLEMENTATION AND TESTING

GENERIC IMPLEMENTATION ISSUES

- Design
  - Display size
  - Presentation of information
  - Interaction (controls)
  - Location relative to operator
- Evaluation
  - By regulators, end users, purchasers…
- Standardization

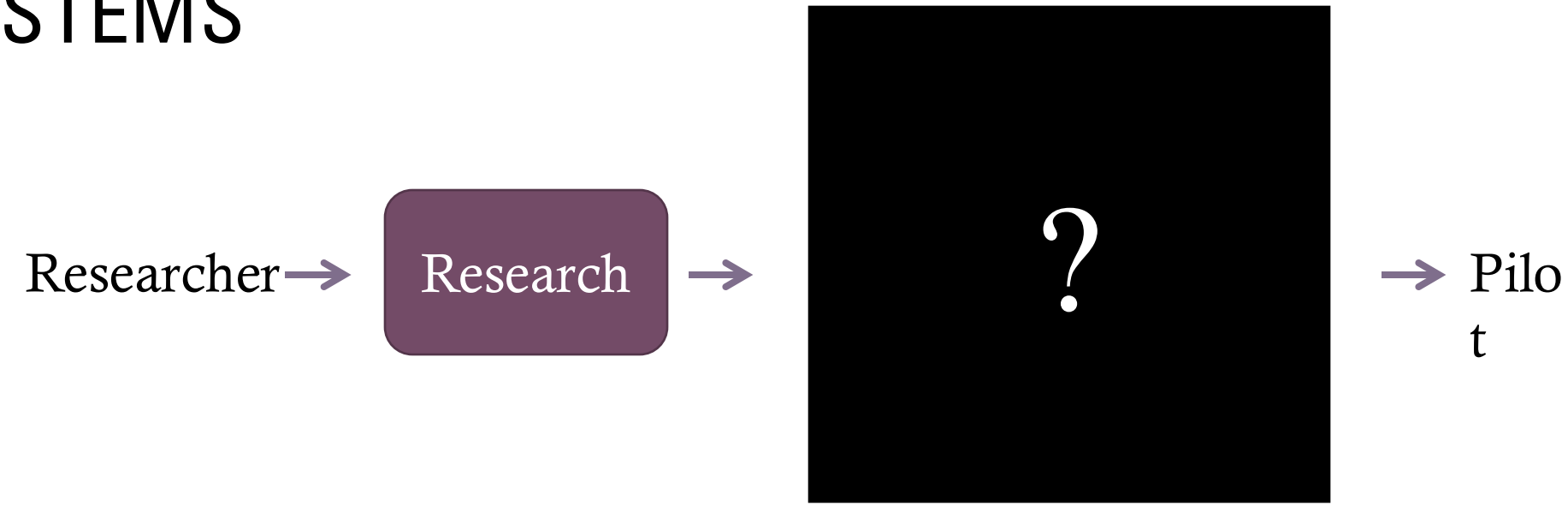
DISPLAY EVALUATION TOPICS AND GUIDANCE

- General issues
  - Legibility of fonts and labels, training, ease of use, “intuitiveness”
- Hardware
  - Screen size
  - Physical controls and interaction
  - Screen technology (e.g., CRT, LCD, HUD)
    - Resolution, refresh rate, viewing angle, brightness (daylight readability)
    - Input devices, e.g., touch screen, cursor control device
- Software
  - Software controls (e.g., buttons, icons)
  - Color
  - Multi-tasking and interaction
    - Data entry, configuration
  - Information time lag (e.g., traffic)
- Design standards and guidance
  - RTCA DO-160 Environmental Testing
  - RTCA DO 178 Software assurance
  - Many others

TECHNIQUES FOR EVALUATING DISPLAYS

- Type of evaluation varies by stage of development, purpose, time/resource limitations
  - Bench test
    - Symbols, interaction, intuitiveness
    - User-interface Assessment checklist
    - Operational evaluation
    - Simulators
    - In-Flight evaluations
  - Inspection
    - Checklist, observation…
- Who is doing the evaluation?
  - Engineer
  - Non-expert
  - (Representative) End user
  - Regulator

A MENTAL MODEL OF RESEARCH ON FLIGHT DECK SYSTEMS



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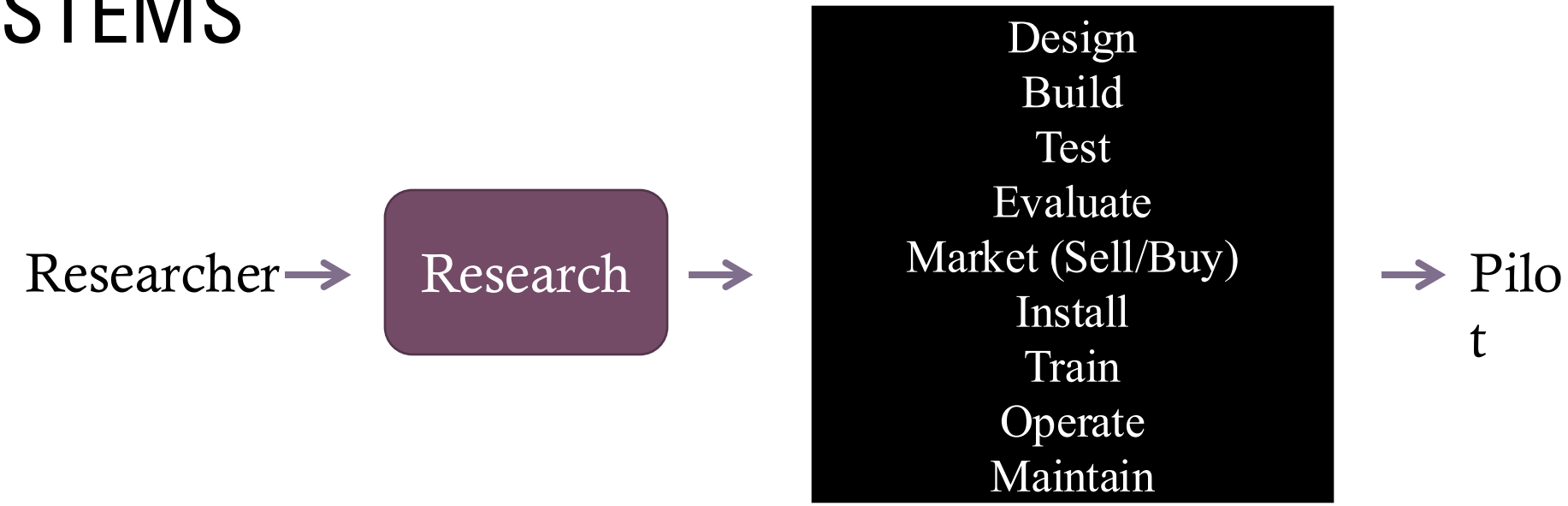

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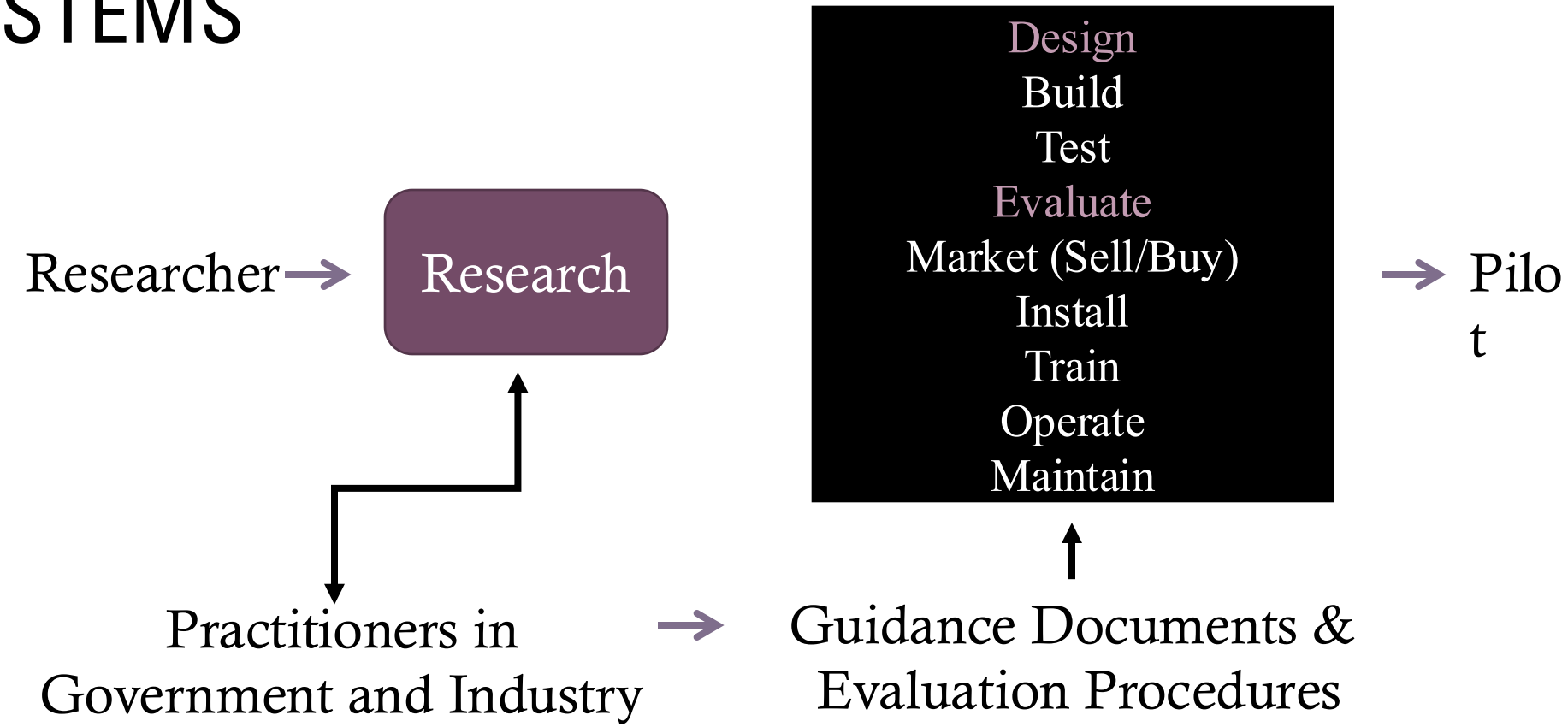

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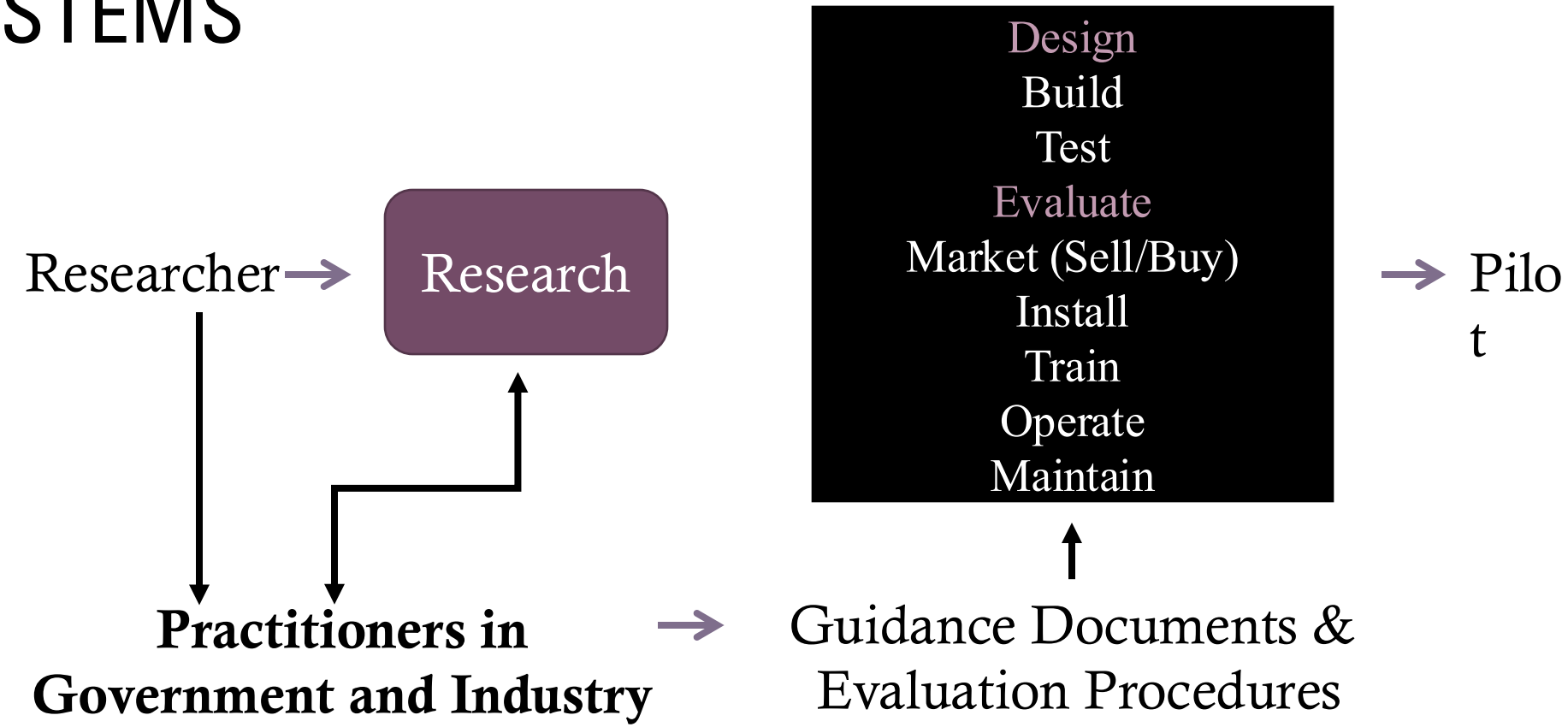

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EXTRAS

MORE ABOUT VISUAL DISPLAY DESIGN

PRINCIPLE OF INFORMATION NEED

Present only the information necessary to carry out the required tasks.

(Need to know what are the required tasks)

767-424 flight deck

PROXIMITY COMPATIBILITY

Effort required for task is decreased when displays that are related, compared, or integrated to perform a task are closer together

Vertical control

Lateral control

User name:

Password:

Sign in

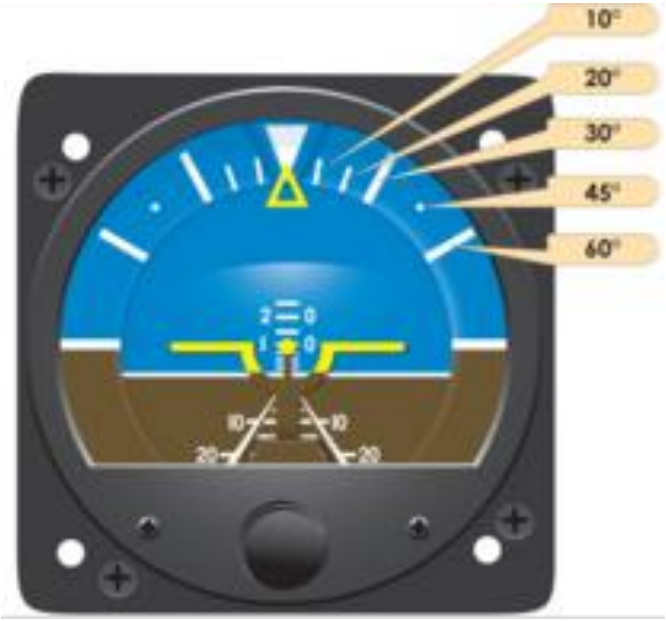
REDUNDANCY

Repeat the message in alternate forms

commons.wikimeida.org

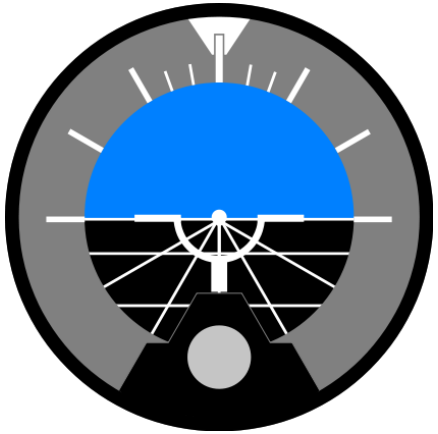
PRINCIPLE OF PICTORIAL REALISM

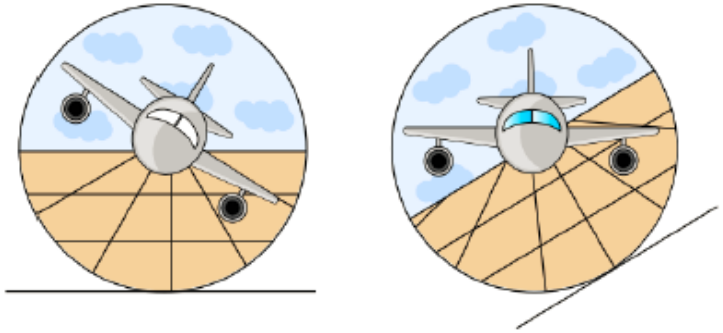
A display should match the human's internal model of the variable that it represents

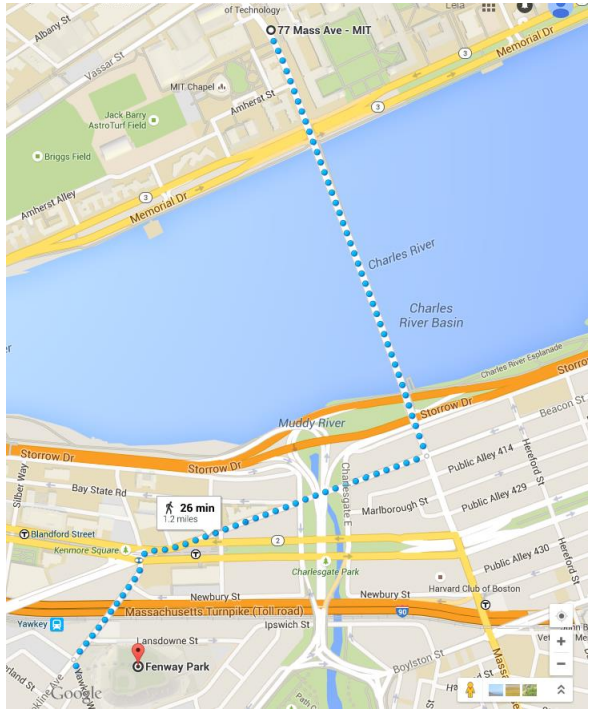


PRINCIPLE OF THE MOVING PART

The moving element on a display should correspond with the element that moves in the user's mental model





PREDICTIVE AIDS

Helps operators to see the future state of the system